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Roll No

IT-5004-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Communication

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Write benefits of digital communication over analog communication. Differentiate bit rate, band rate and bandwidth along with proper example.
b) What is the need of Modulation in digital communication? Briefly discuss few methods to do so.
2. a) Draw block diagram of PCM system and explain its working. Define intersymbol interference and eye-patterns.
b) 24 telephone channels, each band limited to 3.4 KHz are to be time division multiplexed by using PCM. Calculate the bandwidth of the PCM system for 128 quantization levels and an 8KHz sampling frequency.
3. a) Explain the working of a BPSK system.
b) Explain the concept of M-Ary FSK system and also explain its working.
4. a) With the help of block diagram explain the working of differential PSK system.
b) Explain the working of Quadrature amplitude modulation.

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5. a) Discuss Shannon's theorem.
b) Distinguish between irreducible and separable codes. What is coding Efficiency?
6. a) Apply Shannon-Fano coding procedure for following data and take $M = 2$.
 $[X] = [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8]$
 $[P] = [0.1 \quad 0.25 \quad 0.15 \quad 0.05 \quad 0.15 \quad 0.1 \quad 0.05 \quad 0.15]$
b) The generator matrix for a (6,3) block code is given below. Find all code vectors of this code.

$$G = \begin{bmatrix} 100 & : & 110 \\ 010 & : & 011 \\ 001 & : & 111 \end{bmatrix}$$

7. a) With the help of suitable figures explain Simplex, half duplex and full duplex transmission.
b) Draw and explain Following line codes
i) Unipolar NRZ ii) Bipolar NRZ
iii) Unipolar RZ iv) Bipolar RZ
8. a) Describe the relationship between simplex, half duplex and full duplex. What role do they play?
b) A (6, 3) linear block code is generated according to the following generator matrix:

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

For a particular code word transmitted, the received code word is 100011. Find the corresponding data word transmitted.
